

ELECTRONIC CALCULATORS MUST NOT BE USED IN THIS PAPER

1 Work out.

(a) 0.05×0.3

0.015

[1]

(b) $600 \div 0.2$

3000

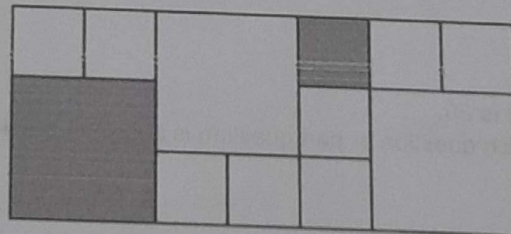
[1]

(c) $20 - 12 \div (8 - 6)$

14

[1]

2



This rectangle is split into squares of two different sizes.

Find the fraction of the rectangle that is shaded grey.

5/21

[1]

- 3 (a) Find the decimal which is exactly halfway between $\frac{3}{5}$ and 68%.

0.64 [1]

- (b) Write 4.073 82 correct to 3 decimal places.

4.074 [1]

- (c) Evaluate $\sqrt[3]{64}$.

4 [1]

- 4 Sonu records the temperature, in $^{\circ}\text{C}$, at midnight every day for 12 days. Here are the results in order, starting with the coldest.

-6 -5 -3 -2 -1 -1 T 5 5 6 6 7

- (a) Find the range of the temperatures.

$$7 - (-6) \quad | \quad \square - \square$$

$$13$$

13 $^{\circ}\text{C}$ [1]

- (b) The median temperature is 1°C .

Find the value of T .

$$\frac{12 + 1}{2} = 6.5$$

$$\frac{-1 + T}{2} = 1$$

$$T = 3$$

$$\frac{\square + \square}{\square} = \square$$

$$\frac{\square + \square}{\square} = 1$$

$$T = \square$$

$T = 3$ [1]

- 5 Anna and Ria share some money in the ratio 5 : 9.
Ria receives \$8 more than Anna.

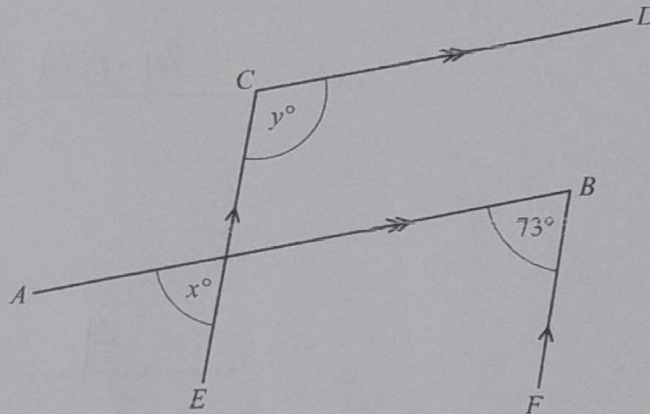
Work out the total amount of money that is shared.

$$\begin{aligned}\frac{x}{x+8} &= \frac{5}{9} \\ 9x &= 5x + 40 \\ 4x &= 40 \\ x &= 10\end{aligned}$$

$$\begin{aligned}\text{Anna} &= 10 \\ \text{Ria} &= 10 + 8 = 18 \\ \text{Total}\end{aligned}$$

\$ 28 [2]

6



NOT TO
SCALE

AB and CD are parallel lines.
 EC and FB are parallel lines.
Angle $ABF = 73^\circ$.

- (a) Find the value of x .

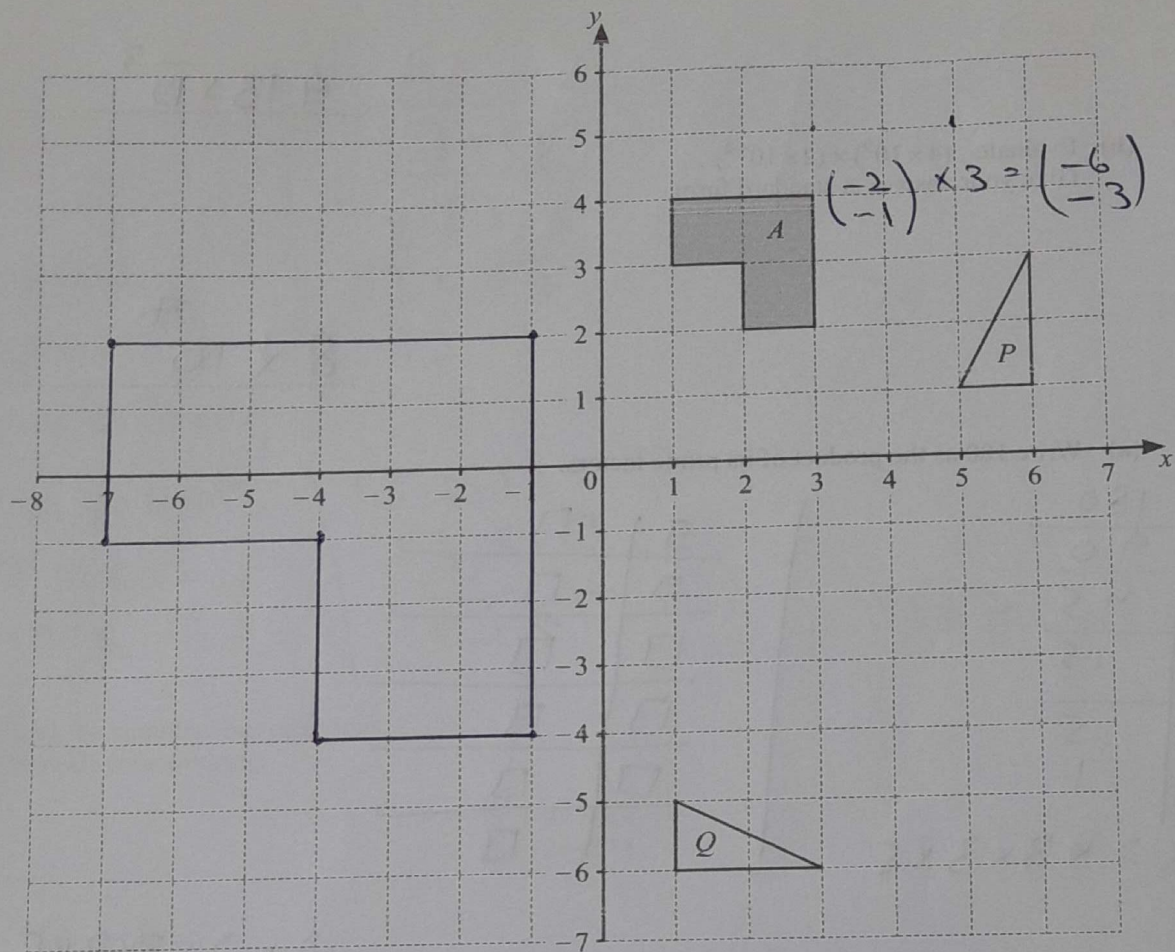
$x = 73$ [1]

- (b) Find the value of y .

$$\begin{aligned}y &= 180 - 73 \\ y &= 107\end{aligned}$$

$y = 107$ [1]

- 7 Shape A and triangles P and Q are drawn on a centimetre square grid.



- (a) Describe fully the **single** transformation that maps triangle P onto triangle Q .

Rotation, 90° clock wise about the centre $(0, 0)$

[3]

- (b) Shape B is an enlargement of shape A .
The centre of enlargement is $(5, 5)$.
The area of shape B is 27 cm^2 .

Draw shape B on the grid.

$$\text{S.F.} = \frac{\text{Area of enlarged shape}}{\text{Area of original shape}}$$

$$\text{S.F.} = \frac{27}{3^2} = 3$$

$$\text{S.F.} = \frac{\square}{\square}$$

$$\text{S.F.} = \square$$

[3]

- 8 (a) Write the number 0.00493 in standard form.

$$4.93 \times 10^{-3} \quad [1]$$

- (b) Evaluate $(4 \times 10^9) \times (2 \times 10^{-2})$.
Give your answer in standard form.

$$8 \times 10^7 \quad [1]$$

- 9 (a) Write 180 as the product of its prime factors.

$$\begin{array}{r|l} 2 & 180 \\ \hline 2 & 90 \\ 3 & 45 \\ 3 & 15 \\ 5 & 5 \\ & 1 \end{array}$$

$$2 \times 2 \times 3 \times 3 \times 5$$

☐	☐
☐	☐
☐	☐
☐	☐
☐	☐
☐	☐
☐	☐

$$2 \times 2 \times 3 \times 3 \times 5 \quad [2]$$

- (b) Expressed as the product of their prime factors,

$$36 = 2^2 \times 3^2 \quad \text{and} \quad N = 2^2 \times 3 \times k, \quad \text{where } k > 3.$$

180 is the lowest common multiple (LCM) of 36 and N .

Find the value of k .

$$k = 5 \quad [1]$$

- 10 By writing each number correct to 1 significant figure, estimate the value of

$$\sqrt{\frac{1240 \times 3.8}{11.2}}$$

$$\sqrt{\frac{1000 \times 4}{10}}$$

$$10 \times 2$$

$$20$$

$$\sqrt{\frac{\square \times \square}{\square}}$$

$$\square \times \square$$

$$\square$$

$$20$$

[2]

- 11 Solve $7m - 13 \leq 8$.

$$7m \leq 8 + 13$$

$$7m \leq 21$$

$$m \leq 3$$

$$\square \leq \square + \square$$

$$\square \leq \square$$

$$\square \leq \square$$

$$m \leq 3$$

[2]

- 12 Solve the simultaneous equations.
Show all your working.

$$5x + 4y = 14$$

$$3x - 2y = 15$$

$$x = 4$$

$$y = -3/2$$

[3]

- 13 A list of eight numbers has a mean of 12.
The first five numbers have a mean of 9.

Find the sum of the three remaining numbers.

$$12 \times 8 = 96$$

$$9 \times 5 = 45$$

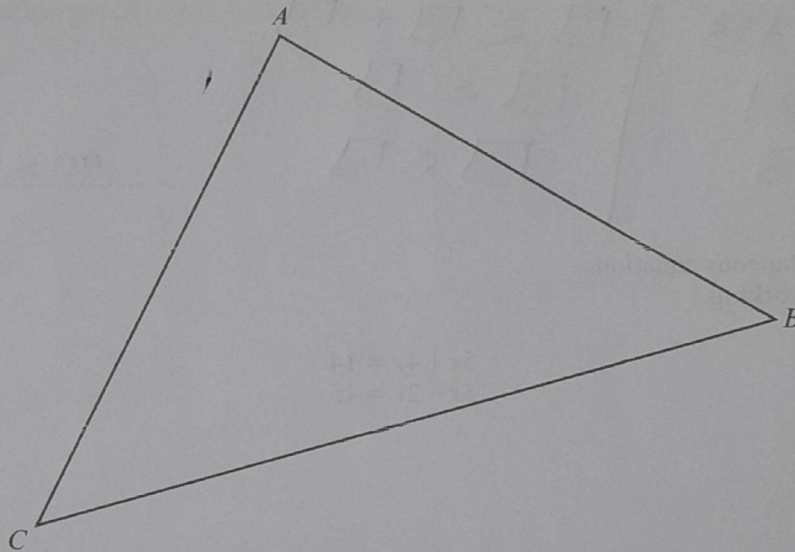
$$96 - 45 = 51$$

$$\begin{array}{l} \square \times 18 = \square \\ \square \times 15 = \square \\ \square - \square = \square \end{array}$$

51

[2]

14



- (a) Measure angle ABC .

Angle $ABC = 47$ to 51 [1]

- (b) Using compasses and a straight edge only, construct the perpendicular bisector of AC . [2]

- (c) On the diagram, shade the region inside triangle ABC that is

- nearer to A than to C
- and
- more than 6 cm from B .

[2]

9

$$d_2 \left| \frac{16 - 28}{3} = -4 \right|$$

- 15 (a) The second term of a linear sequence is 28.
The fifth term of the sequence is 16.

Find the first term, the third term and the fourth term of this sequence.

$$\begin{array}{l|l} a + d = 28 & a + d = 28 \\ a + 4d = 16 & a - 4 = 28 \\ \hline -3d = 12 & a = 32 \\ d = -4 & \end{array}$$

First term = 32

Third term = 24

Fourth term = 20 [2]

- (b) These are the first five terms of a different sequence.

3 9 19 33 51

Find an expression for the n th term of this sequence.

$$\begin{array}{cccccc} 3 & 9 & 19 & 33 & 51 \\ +6 & +10 & +14 & +18 \\ +4 & +4 & +4 \end{array}$$

$$a = 4/2 = 2$$

$$b = 6 - 3(2)$$

$$b = 0$$

$$c = 3 - 2 - 0$$

$$c = 1$$

$$2n^2 + 1$$

[2]

16 $T = \sqrt{P-4}$

- (a) Work out the value of T when $P = 40$.

$T =$ 6 [1]

- (b) Rearrange the formula to make P the subject.

$$T^2 = P - 4$$

$$T^2 + 4 = P$$

$$\square^2 = \square - 13$$

$$\square + 13 = \square$$

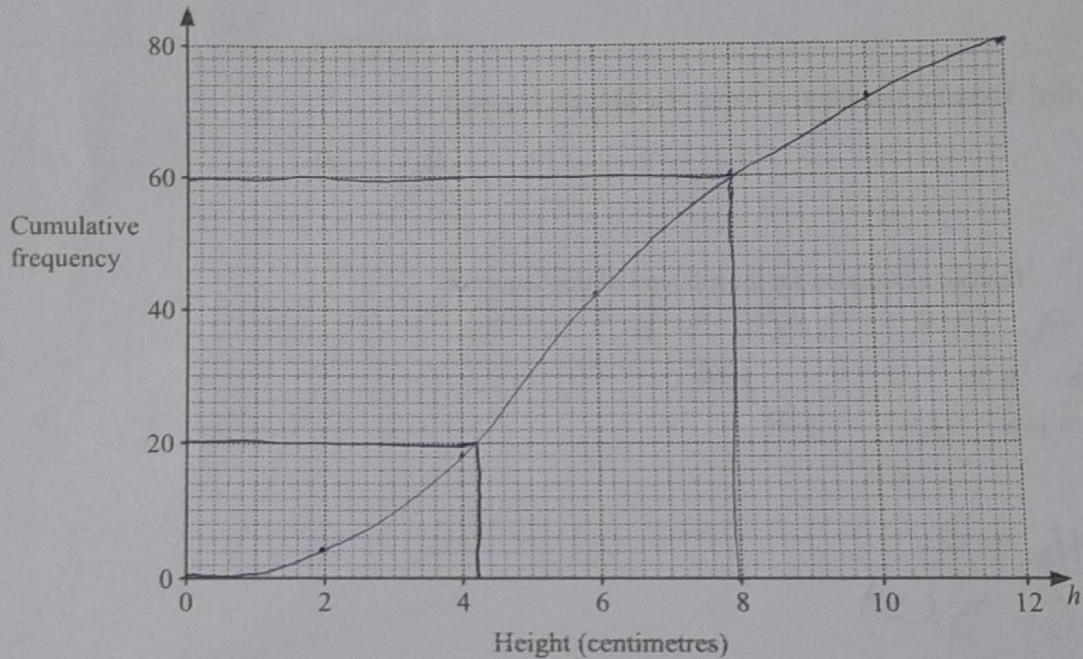
$$T^2 + 4$$

$P =$ $T^2 + 4$ [2]

- 17 The heights of 80 plants are measured.
The table shows the results.

Height (h centimetres)	$h \leq 2$	$h \leq 4$	$h \leq 6$	$h \leq 8$	$h \leq 10$	$h \leq 12$
Cumulative frequency	4	18	42	60	72	80

- (a) Draw a cumulative frequency diagram to show this information.



[2]

- (b) Use your diagram to find an estimate for the interquartile range.

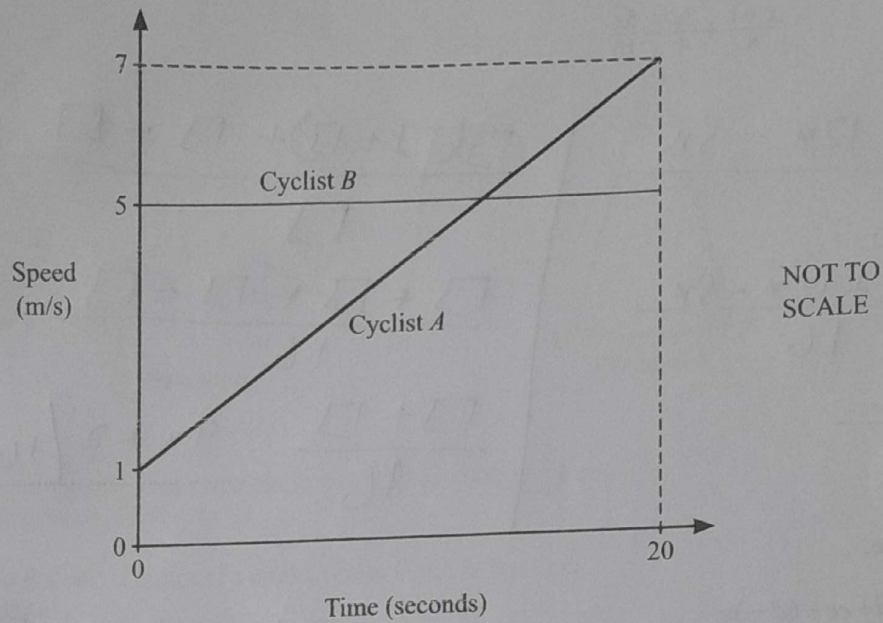
$$\begin{aligned}
 & \text{IQR} = Q_3 - Q_1 \quad | \quad \text{IQR} = 60 - 20 \quad | \quad \text{IQR} = 8 - 4.2 \\
 & Q_1 = \frac{1}{4} \times 80 = 20 \quad | \quad \text{IQR} = 40 \quad | \quad \text{IQR} = 3.8 \text{ cm} \quad [2] \\
 & Q_3 = \frac{3}{4} \times 80 = 60
 \end{aligned}$$

- (c) Plants are sold when they are taller than H centimetres.
28 of these plants are sold.

Find the value of H .

$$\begin{aligned}
 & 80 - H = 28 \quad | \quad \square - H = \square \quad | \quad H = 52 \quad [2] \\
 & H = 52 \quad | \quad H = \square
 \end{aligned}$$

- 18 The diagram shows the speed-time graph of part of a journey for two cyclists, A and B.



- (a) Find the acceleration of cyclist A during the first 20 seconds.

$$6/20 = 3/10 = 0.3$$

$$0.3 \text{ m/s}^2 [1]$$

- (b) Find which cyclist travelled further in the first 20 seconds and by how many metres.

Distance travelled by A:

$$S_A = \frac{1}{2} (1 + 7) \times 20 = 80 \text{ m}$$

Distance travelled by B:

$$S_B = 5 \times 20 = 100 \text{ m}$$

$$S_A = \frac{1}{2} (13 + 13) \times 13 = 13$$

$$S_B = 13 \times 13 = 13 \text{ m}$$

Cyclist B travelled further by 20 metres [3]

19 Express as a single fraction in its simplest form.

$$\frac{x+1}{8} + \frac{3x}{4} - \frac{5x}{16}$$

$$\frac{2(n+1) + 12n - 5n}{16}$$

$$\frac{2n + 2 + 12n - 5n}{16}$$

$$\frac{9n + 2}{16}$$

$$\frac{\square(\square + \square) + \square - \square}{16}$$

$$\frac{\square + \square + \square - \square}{16}$$

$$\frac{\square + \square}{16} \quad 9n + 2 / 16 \quad [2]$$

20 Factorise.

(a) $2cd + ce - 6d - 3e$

$$c(2d + e) - 3(2d + e)$$

$$(c - 3)(2d + e)$$

$$\frac{\square(\square + \square) - \square(\square + \square)}{(\square - \square)(\square + \square)}$$

$$(c - 3)(2d + e) \quad [2]$$

(b) $3v^2 - 27t^2$

$$3(v^2 - 9t^2)$$

$$3(v + 3t)(v - 3t)$$

$$\square(\square^2 - \square^2)$$

$$\square(\square + \square)(\square - \square)$$

$$3(v + 3t)(v - 3t) \quad [2]$$

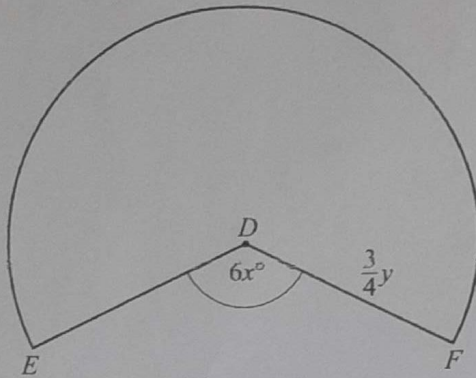


Diagram A

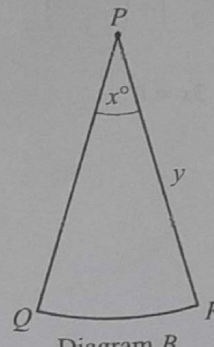


Diagram B

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Diagram A shows a sector of a circle, centre D and radius $\frac{3}{4}y$ cm.
The obtuse angle $EDF = 6x^\circ$.

Diagram B shows a sector of a circle, centre P and radius y cm.
The sector angle is x° .

- (a) The length of the major arc EF is 9 times the length of the arc QR .

Show that $x = 20$.

$$\frac{360 - 6x}{360} \times 2\pi \times \frac{3y}{4} = 9 \times \frac{x}{360} \times 2\pi y \quad \left| \quad \frac{360 - 6x}{360} \times 2\pi \times \frac{3y}{4} = 9 \times \frac{x}{360} \times 2\pi y \right.$$

$$360 - 6x \times \frac{3y}{4} = \frac{9x}{4} y$$

$$360 - 6x = 12x$$

$$360 = 18x$$

$$\frac{360}{18} = x$$

$$x = 20$$

$$x = 20$$

[3]

- (b) Find the value of y when the area of sector QPR is equal to $2\pi \text{ cm}^2$.

$$A = \frac{x}{360} \times \pi (y)^2 = 2\pi$$

$$2\pi = \frac{x}{360} \times \pi y^2$$

$$2 = \frac{x}{360} \times y^2$$

$$y^2 = 36$$

$$y = 6$$

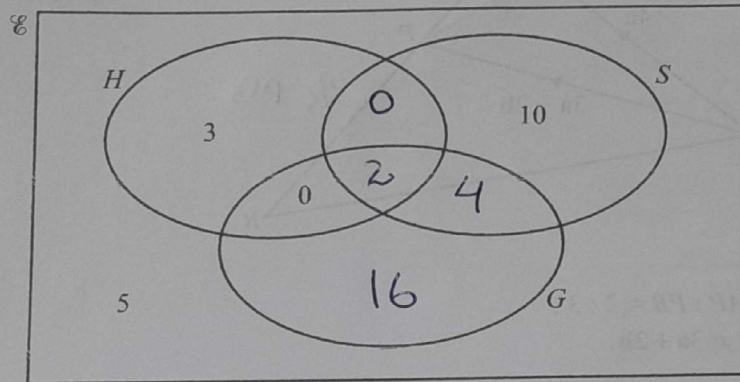
$$A = \frac{x}{360} \times \pi (y)^2$$

$$2\pi = \frac{x}{360} \times \pi (y)^2$$

$$y = 6$$

[2]

- 23 A shop sells hats (H), scarves (S) and gloves (G).
A group of 40 people are asked which items they buy in the shop.
Some of the results are shown in the Venn diagram.



- (a) 2 people buy all three items.
Those people that buy both a hat and a scarf also buy gloves.
4 people buy exactly two items.

Use this information to complete the Venn diagram.

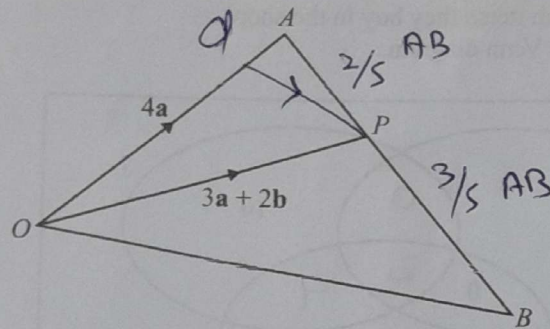
[2]

- (b) Work out $n(S \cap (H \cup G)')$.

10

[1]

Question 24 is printed on the next page.



NOT TO SCALE

OAB is a triangle.

P lies on AB and $AP : PB = 2 : 3$.

$\vec{OA} = 4\mathbf{a}$ and $\vec{OP} = 3\mathbf{a} + 2\mathbf{b}$.

(a) Find, in terms of $\mathbf{a}$ and $\mathbf{b}$, giving your answer in its simplest form

(i) $\vec{AP}$

$$\begin{aligned}\vec{AP} &= \vec{AO} + \vec{OP} \\ \vec{AP} &= -4\mathbf{a} + 3\mathbf{a} + 2\mathbf{b} \\ \vec{AP} &= 2\mathbf{b} - \mathbf{a}\end{aligned}$$

$$\begin{aligned}\vec{AP} &= \square + \square \\ \vec{AP} &= \square + \square + \square \\ \vec{AP} &= \square - \square\end{aligned}$$

$$\vec{AP} = 2\mathbf{b} - \mathbf{a} \quad [1]$$

(ii) $\vec{OB}$

$$AP = \frac{2}{5} \vec{AB}$$

$$2\mathbf{b} - \mathbf{a} = \frac{2}{5} \vec{AB}$$

$$\vec{AB} = 5\mathbf{b} - \frac{5}{2}\mathbf{a}$$

$$\begin{aligned}\vec{OB} &= \vec{OA} + \vec{AB} \\ &= 4\mathbf{a} + 5\mathbf{b} - \frac{5}{2}\mathbf{a} \\ &= \frac{8\mathbf{a} + 10\mathbf{b} - 5\mathbf{a}}{2} \\ &= \frac{3\mathbf{a} + 10\mathbf{b}}{2}\end{aligned}$$

$$\vec{OB} = \frac{3\mathbf{a} + 10\mathbf{b}}{2} \quad [3]$$

(b) Q is a point on OA such that $\vec{QP}$ is parallel to $\vec{OB}$.

Find $\vec{QP}$.

$$\vec{QP} = k \vec{OB}$$

$$\vec{QP} = \frac{3}{5}\mathbf{a} + 2\mathbf{b} \quad [1]$$

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